(4) Quotients of polynomials namely rational functions of the form

$R(x) = \dfrac{p(x)}{q(x)}$, are continuous at every point where $q(x) \neq 0$, and

$$\lim_{x \to x_0} R(x) = \lim_{x \to x_0} \frac{p(x)}{q(x)}$$

$$= \frac{\lim\limits_{x \to x_0} p(x)}{\lim\limits_{x \to x_0} q(x)} = \frac{p(x_0)}{q(x_0)} = R(x_0) .$$

(5) The circular functions sin x and cos x are continuous at every point of their domain $\mathbb{R} = (-\infty, \infty)$ since $\lim\limits_{x \to x_0} \sin x = \sin x_0$, $\lim\limits_{x \to x_0} \cos x = \cos x_0$.

As a consequence, tan x, cot x, cosec x, sec x are continuous on their proper domains in view of the reciprocal and quotient rules in the algebra of limits.

(6) The nth root functions, $f(x) = x^{\frac{1}{n}}$ are continuous in their proper domain since $\lim\limits_{x \to x_0} \left(x^{\frac{1}{n}} \right) = x_0^{\frac{1}{n}}$.

(7) The reciprocal function $f(x) = \dfrac{1}{x}$ is not defined at 0 and hence it is not continuous at 0. It is continuous at each point of $\mathbb{R} - \{0\}$.

(8) $h(x) = \begin{cases} x+1, & x \leq 0 \\ x^2+1, & x > 0 \end{cases}$

The domain of h is all of real numbers and

$$\lim_{x \to 0^-} h(x) = \lim_{x \to 0^-} (x+1) = 1 = h(0)$$

$$\lim_{x \to 0^+} h(x) = \lim_{x \to 0^+} (x^2+1) = 1 = h(0) .$$

Thus $h(x)$ is continuous at $x = 0$.

Indeed, $h(x)$ is continuous at each point of $(-\infty, 0)$ and each point of $(0, \infty)$ and hence h is continuous in the whole of $(-\infty, \infty)$.

(9) The greatest integer function $f(x) = \lfloor x \rfloor$ is not continuous at $x = 0$.

For, $\lim\limits_{x \to 0^-} \lfloor x \rfloor = -1$ and

$\lim\limits_{x \to 0^+} \lfloor x \rfloor = 0$

It is discontinuous at each integer point. In fact,

$\lim\limits_{x \to n^-} \lfloor x \rfloor = n - 1$ and

$\lim\limits_{x \to n^+} \lfloor x \rfloor = n$.

09_11th-Maths-Vol2-Unit9.indd 122 16-12-2022 11:03:45